

1. Textbook problems.

- 1.2 Calculate the altitude at which an artificial satellite orbiting in the equatorial plane can be a synchronous satellite (i.e., can remain above the same spot on the surface of Earth).

Ans:

Gravity provides centripetal acceleration. $\vec{F}_g = \vec{F}_c \rightarrow \vec{g} = \vec{a}_c$

From text Eq. (1.7)

$$g^* = -\frac{GM}{r^2} \left(\frac{\vec{r}}{r} \right), \vec{a}_c = \Omega^2 R$$

and at the equator $R = r$. Thus,

$$\Omega^2 r = -g^* \Rightarrow r^3 = \frac{GM}{\Omega^2}$$

$$r = \sqrt[3]{\frac{GM}{\Omega^2}} \approx 42200 \text{ km}$$

Then $z = r - a \approx 35,830 \text{ km}$.

- 1.6 Two balls 4 cm in diameter are placed 100 m apart on a frictionless horizontal plane at 43N. If the balls are impulsively propelled directly at each other with equal speeds, at what speed must they travel so that they just miss each other?

Ans:

Assume that the ball is thrown in speed of $u = x/t$, where $x = 100 \text{ m}$ and t is the time.

From Eq(1.21) $v = -fu_0t$

$$v = \frac{dy}{dt} = -(2\Omega \sin \phi)ut \Rightarrow |y| = (2\Omega \sin \phi)xt/2$$

$$t = \frac{2|y|}{(2\Omega \sin \phi)x} = \frac{2 \times 2 \times 10^{-2}}{(2 \times 7.292 \times 10^{-5} \times \sin \phi) \times 50} = 8s$$

$$u \frac{x}{t} = \frac{50m}{8s} = 6.25s$$

- 1.8 Find the horizontal displacement of a body dropped from a fixed platform at a height h at the equator, neglecting the effects of air resistance. What is the numerical value of the displacement for $h = 5 \text{ km}$?

Ans:

From the Eq.(1.16)

$$\frac{du}{dt} = (2\Omega \sin \phi)v - (2\Omega \cos \phi)w - \frac{uw}{r} + \frac{uv}{r} \tan \phi$$

$$u, v, w = 0 \text{ and } \phi = 0 \Rightarrow \left(\frac{du}{dt} \right)_{Co} = -2\Omega w \text{ at the equator. Since}$$

$$\frac{dw}{dt} \approx -g \rightarrow \frac{dz}{dt} = -gt \rightarrow z = \frac{gt^2}{2}$$

time of free fall from height h , $t_0 = \sqrt{\frac{2h}{g}}$ we get

$$\frac{du}{dt} = 2\Omega gt \Rightarrow u = \frac{dx}{dt} = \Omega gt^2 \rightarrow x = \Omega gt^3/3$$

$$x = \frac{\Omega g}{3} \left(\frac{2h}{g} \right)^{\frac{3}{2}} = 7.8 \text{ m eastward for } h = 5 \text{ km}$$

- 1.9 A bullet is fired directly upward with initial speed ω_0 at latitude ϕ . Neglecting air resistance, by what distance will it be displaced horizontally when it returns to the ground? (Neglect $2u \cos \phi$ compared to g in the vertical momentum equation.)

Duo to gravity, $\frac{dw}{dt} = -g$, so that $w = w_0 - gt = 0$

Total time of flight is twice the time to the top, or $t_0 = \frac{2w_0}{g}$

From Equ.(1.16), and only consider w

$\frac{du}{dt} = -(2\Omega \cos \phi)w$, which integrates to give

$$u = -(2\Omega \cos \phi)w(w_0t - gt^2/2)$$

Since $\frac{dx}{dt} = u$

$$x = 2\Omega \cos \phi(w_0t^2/2 - gt^3/6)$$

Substituting the total time t_0 , then give

$$x = -2\Omega \cos \phi = (2w_0^3/3g^2)$$

- 1.11 A particle is free to slide on a horizontal frictionless plane located at a latitude ϕ on Earth. Find the equation governing the path of the particle if it is given an impulsive northward velocity $v = V_0$ at $t = 0$. Give the solution for the position of the particle as a function of time. (Assume that the latitudinal excursion is sufficiently small that f is constant.)

Ans:

From motion equation of Coriolis motion:

$$\begin{cases} \frac{du}{dt} = 2\Omega v \sin \phi = fv \\ \frac{dv}{dt} = -2\Omega u \sin \phi = -fu \end{cases}$$

we combine the eq. then

$$\frac{d}{dt}\left(\frac{dv}{dt}\right) = -f\left(\frac{du}{dt}\right) = -f^2v$$

Let the general solution is $v = A \cos ft + B \sin ft$,

then $v' = -Af \sin ft + Bf \cos ft$

Substitute the initial conditions that $v = V_0$ at $t = 0$

we get $A = V_0$ and $B = 0$

Therefore,

$$\frac{dy}{dt} = v = V_0 \cos ft$$

Do derivative to v, then we can get u

$$\frac{dv}{dt} = u = V_0 \sin ft$$

Integrating with respect to time

$$\begin{aligned} \int \frac{dy}{dt} &= \int_0^t V_0 \cos ft = \frac{1}{f} V_0 \sin ft \\ \int \frac{dx}{dt} &= \int_0^t V_0 \sin ft = -\frac{1}{f} V_0 \cos ft + \frac{V_0}{f} \end{aligned}$$

Letting (x_0, y_0) be the initial position

$$\begin{cases} x = x_0 - \frac{V_0}{f} (\cos ft - 1) \\ y = y_0 + \frac{V_0}{f} \sin ft \end{cases}$$

- 1.13 Isolines of 1000 to 500 hPa thickness are drawn on a weather map using a contour interval of 60 m. What is the corresponding layer mean temperature interval?

Ans:

According to Hydrostatic balance Eq.

$$Z_T = \frac{R\bar{T}}{g_0} \ln\left(\frac{P_1}{P_2}\right)$$

Thus,

$$\bar{T} = \frac{Z_T g_0}{R \ln(P_1/P_2)} = \frac{60 \times 9.8}{287 \times \ln 2} = 2.96 \approx 3^\circ\text{C}$$

- 1.14 Show that a homogeneous atmosphere (density independent of height) has a finite height that depends only on the temperature at the lower boundary. Compute the height of a homogeneous atmosphere with surface temperature $T_0 = 273K$ and surface pressure 1000 hPa. (Use the ideal gas law and hydrostatic balance.)

Ans:

According to hydrostatic balance assumption, $\frac{dp}{dz} = -\rho g = -\rho_0 g$ as $\rho = \text{Constant}$
integrate both side

$$\begin{aligned}\int_{P_0}^0 dp &= -\rho_0 g \int_0^H dz \\ P_0 &= \rho_0 g H = \rho_0 R T_0 \\ H &= \frac{p_0}{\rho_0 g} = \frac{R T_0}{g} = \frac{287 \times 273}{9.8} \approx 7.99 \text{ km}\end{aligned}$$

- 1.16 Show that in an atmosphere with uniform *lapserate* γ (where $\gamma \equiv -dT/dz$) the geopotential height at pressure level p_1 is given by

$$Z = \frac{T_0}{\gamma} \left[1 - \left(\frac{p_0}{p_1} \right)^{-R\gamma/g} \right]$$

where T_0 and p_0 are the sea-level temperature and pressure, respectively.

Ans:

According to definition of geopotential:

$$d\Phi = g dZ = -(RT/p) dp = -RT d \ln p$$

$$d \ln p = -\frac{g_0}{RT} dZ = -\frac{g_0}{R} \frac{dZ}{(T_0 - \gamma Z)}$$

integrate both side

$$\begin{aligned}\int_{p_0}^{p_1} d \ln p &= -\frac{g_0}{R} \int_0^Z \frac{dZ}{(T_0 - \gamma Z)} \\ \ln \left(\frac{p_1}{p_0} \right) &= \frac{g_0}{R\gamma} \ln \left(\frac{T_0 - \gamma Z}{T_0} \right) \\ \left(\frac{p_1}{p_0} \right) &= \left(\frac{T_0 - \gamma Z}{T_0} \right)^{\frac{g_0}{R\gamma}}\end{aligned}$$

reorganize the equation to the following form

$$Z = \frac{T_0}{\gamma} \left[1 - \left(\frac{p_0}{p_1} \right)^{-R\gamma/g} \right]$$

- 2.1 A ship is steaming northward at a rate of 10 km h^{-1} . The surface pressure increases toward the northwest at the rate of 5 Pa km^{-1} . What is the pressure tendency recorded at a nearby island station if the pressure aboard the ship decreases at a rate of 100 Pa / 3h ?

Ans:

$$\begin{aligned}\frac{\partial P}{\partial t} &= \frac{dp}{dt} - u \cdot \nabla p \\ u \cdot \nabla p &= |u| |\nabla p| \cos \alpha = u \cdot \frac{\partial p}{\partial x} \cos \alpha\end{aligned}$$

where $\alpha = 45^\circ$ is the angle between the velocity and pressure gradient vector.

$$\frac{\partial P}{\partial t} = -\left(\frac{100}{3} \text{ Pa/h}\right) - (10 \text{ km/h})(5 \text{ Pa/km}) \frac{1}{\sqrt{2}} = -68.7 \text{ Pa/h}$$

- 2.2 The temperature at a point 50 km north of a station is 3°C cooler than at the station. If the wind is blowing from the northeast at 20 m s^{-1} and the air is being heated by radiation at the rate of 1°C h^{-1} , what is the local temperature change at the station?

Ans:

$$\frac{\partial T}{\partial t} = \frac{dT}{dt} - v \cdot \nabla T$$

$$\frac{dT}{dt} = 1^\circ\text{C}^{-1}$$

$$v \cdot \nabla T = (20\text{ms}^{-1})(3^\circ\text{C}/5 \times 10^4\text{m})(1/\sqrt{2}) = 8.47 \times 10^{-4}^\circ\text{C}^{-1} = 3.05^\circ\text{C}^{-1}$$

Thus,

$$\frac{\partial T}{\partial t} = 1^\circ\text{C}/h - 3.05^\circ\text{C}/h = 2.05^\circ\text{C}/h$$

- 2.5 Suppose a 1-kg parcel of dry air is rising at a constant vertical velocity. If the parcel is being heated by radiation at the rate of 10^{-1}Wkg^{-1} , what must the speed of rise be to maintain the parcel at a constant temperature?

Ans:

From first law of thermodynamics:

$$C_p \frac{dT}{dt} - \alpha \frac{dp}{dt} = J$$

Substitute αdp into $-gdz$ by the hydrostatic equation. Therefore,

$$C_p \frac{dT}{dt} - \alpha \frac{dp}{dt} = C_p \frac{dT}{dt} + gw = J$$

for the parcel at a constant temperature. ($dT/dt = 0$)

$$w = \frac{J}{g} = (10^{-1} \text{Wkg}^{-1})/(9.8 \text{ms}^{-2}) = 0.0102 \text{ms}^{-1}$$

- 2.7 An air parcel that has a temperature of 20°C at the 1000-hPa level is lifted dry adiabatically. What is its density when it reaches the 500-hPa level?

Ans:

For adiabatic expansion potential temperature is conserved. Thus,

$$\theta = T_s = T \left(\frac{p_s}{p} \right)^{\frac{R}{C_p}}$$

Substituting from the ideal gas law, $T = \frac{p}{\rho R}$

$$RT_s = \frac{p}{\rho} \left(\frac{p_s}{p} \right)^{\frac{R}{C_p}} = \frac{p_s}{\rho_s} \implies \rho = \rho_s \left(\frac{p}{p_s} \right)^{C_v/C_p}$$

where we have used $C_v = C_p - R$.

$$\rho(500\text{hPa}) = \left[\frac{10^5 \text{Pa}}{(287 \text{JK}^{-1}\text{kg}^{-1})(293\text{K})} \right] \left(\frac{1}{2} \right)^{0.714} = 0.725 \text{kgm}^{-3}$$

- 2.8 Suppose an air parcel starts from rest at the 800-hPa level and rises vertically to 500 hPa while maintaining a constant 1°C temperature excess over the environment. Assuming that the mean temperature of the 800- to 500- hPa layer is 260 K, compute the energy released due to the work of the buoyancy force. Assuming that all the released energy is realized as kinetic energy of the parcel, what will the vertical velocity of the parcel be at 500 hPa?

Ans:

According to Static Stability, The vertical acceleration of the parcel is

$$\frac{d^2}{dt^2}(\delta z) = g \left(\frac{\rho_0 - \rho}{\rho} \right) = g \frac{\theta}{\theta_0}$$

Substituting from the ideal gas law $\rho = p/RT$ into

$$\frac{dw}{dt} = g \left(\frac{T_0^{-1} - T^{-1}}{T^{-1}} \right) = g \left(\frac{T - T_0}{T_0} \right)$$

where dw/dt is the force per unit mass.

From the hypsometric equation.

$$\delta Z = \frac{RT_0}{g} \ln(p_1/p_2)$$

Energy released is given by force $\times$ distance $W = FS$, thus

$$g\delta Z \left(\frac{T-T_0}{T_0} \right) = R(T-T_0) \ln \frac{p_1}{p_2} = 135 \text{ Jkg}^{-1}$$

If all goes into kinetic energy,

$$K_e = 135 = \frac{w^2}{2} \implies w = 16.43 \text{ ms}^{-1}$$

- 2.11 In the isentropic coordinate system (see Section 4.6), potential temperature is used as the vertical coordinate. Because potential temperature in adiabatic flow is conserved following the motion, isentropic coordinates are useful for tracing the actual paths of travel of individual air parcels. Show that the transformation of the horizontal pressure gradient force from z to θ coordinates is given by

$$\frac{1}{\rho} \nabla_z p = \nabla_\theta M$$

where $M \equiv C_p T + \Phi$ is the Montgomery streamfunction.

Ans:

horizontal distance δx , the pressure difference evaluated along a surface of constant x is

$$\left(\frac{\partial p}{\partial x} \right)_\theta = \left(\frac{\partial p}{\partial z} \right) \left(\frac{\partial z}{\partial x} \right)_\theta + \left(\frac{\partial p}{\partial x} \right)_z$$

Let $\frac{\partial p}{\partial z} = \rho g$, then

$$\begin{aligned} \left(\frac{\partial p}{\partial x} \right)_\theta &= (-\rho g) \left(\frac{\partial z}{\partial x} \right)_\theta + \left(\frac{\partial p}{\partial x} \right)_z \\ \frac{1}{\rho} \left(\frac{\partial p}{\partial x} \right)_z &= \frac{1}{\rho} \left(\frac{\partial p}{\partial x} \right)_\theta + \left(\frac{\partial \Phi}{\partial x} \right)_\theta \end{aligned}$$

where $gdz = d\phi$

With ideal gas law $\frac{1}{\rho} = \frac{RT}{p}$, then

$$\frac{1}{\rho} \left(\frac{\partial p}{\partial x} \right)_\theta = RT \left(\frac{\partial \ln p}{\partial x} \right)_\theta$$

and Eq.(2.44): $\ln \theta = \ln T - \frac{R}{c_p} \ln p + C$, then

$$\left(\frac{\partial \ln p}{\partial x} \right)_\theta = \frac{c_p}{R} RT \left(\frac{\partial \ln T}{\partial x} \right)_\theta$$

Thus,

$$\frac{1}{\rho} \left(\frac{\partial p}{\partial x} \right)_z = \frac{c_p}{T} \left(\frac{\partial \ln T}{\partial x} \right)_\theta + \left(\frac{\partial \Phi}{\partial x} \right)_\theta$$

which simplifies to

$$\begin{aligned} \frac{1}{\rho} \left(\frac{\partial p}{\partial x} \right)_z &= \left(\frac{\partial M}{\partial x} \right)_\theta \\ \frac{1}{\rho} \nabla_z p &= \nabla_\theta M \end{aligned}$$

- 2.12 French scientists have developed a high-altitude balloon that remains at constant potential temperature as it circles Earth. Suppose such a balloon is in the lower equatorial stratosphere where the temperature is isothermal at 200 K. If the balloon were displaced vertically from its equilibrium level by a small distance δz , it would tend to oscillate about the equilibrium level. What would be the period of this oscillation?

The parcel oscillation frequency is

$$N^2 = g \frac{d \ln \theta_0}{dz}$$

Under isothermal conditions $\frac{T}{\theta} \frac{\partial \theta}{\partial z} = \frac{\partial T}{\partial z} + \frac{g}{c_p}$ will be

$$\frac{d \ln \theta_0}{dz} = \frac{g}{c_p T} = \frac{9.8}{1004 \times 200} = 4.885 \times 10^{-5} \text{ m}^{-1}$$

Thus, the period of the oscillation is

$$\frac{2\pi}{N} = \frac{2\pi}{\sqrt{9.8 \times 4.885 \times 10^{-5}}} = 287 \text{ s}$$

2. Inclass problems.

- (1) Suppose an infinite plate moves at a speed of V_0 . The fluid above the plate, stationary at initial time and with thickness z , is only forced by viscous force from the plate below. What will the velocity of the fluid be when it reaches steady state ($t \rightarrow \infty$)? i.e., solve the given boundary-value problem

$$\begin{aligned}\frac{\partial v}{\partial t} &= \nu \frac{\partial^2 v}{\partial z^2} \\ z = 0, v &= V_0 \\ z \rightarrow \infty, v &= 0\end{aligned}$$

Ans: To condense the calculation, assume a unitless number $\xi = \frac{z}{2\sqrt{\nu t}}$

And u is the function of ξ , therefore $u = f(\xi)$

$$\begin{aligned}u_t &= f'(\xi) \frac{\partial \xi}{\partial t} = f'(\xi) \frac{z}{4\sqrt{\nu t^3}} \\ u_{zz} &= \frac{\partial}{\partial z} \left(f'(\xi) \frac{1}{2\sqrt{\nu t}} \right) = \frac{f''(\xi)}{4\nu t}\end{aligned}$$

Substitute into ξ

$$f'(\xi) \frac{z}{4\sqrt{\nu t^3}} = \nu \cdot \frac{f''(\xi)}{4\nu t} \rightarrow f''(\xi) + 2\xi f'(\xi) = 0$$

let $f'(\xi) = g(\xi)$, then

$$g'(\xi) + 2\xi g(\xi) = 0$$

solve the ODE:

$$\begin{aligned}g(\xi) &= ce^{-\xi^2} \\ f(\xi) &= c \int_0^\xi e^{-\xi^2} d\xi + d = v \\ v &= c \int_0^{\frac{z}{2\sqrt{\nu t}}} e^{-\xi^2} d\xi + d\end{aligned}$$

As $z = 0, v = V_0 \implies d = V_0$

As $z = \infty, v = 0$, thus

$$V_0 + c \int_0^\infty e^{-\xi^2} d\xi = 0 \implies V_0 + \frac{\sqrt{\pi}}{2} c = 0 \implies c = \frac{-2V_0}{\sqrt{\pi}}$$

Result:

$$v = V_0 + \frac{-2V_0}{\sqrt{\pi}} \int_0^{\frac{z}{2\sqrt{\nu t}}} e^{-\xi^2} d\xi$$

- (3) Show that the apparent forces on earth in x-dimension resulted from vertical motion ($w \neq 0$) is given by

$$\frac{du}{dt} = -2\Omega w \cos \phi - \frac{uw}{a}$$

If there's no external force, angular momentum will be conservation $\frac{dL}{dt} = 0$

$$L = \Omega R^2 + uR$$

$$\frac{dL}{dt} = \frac{\partial L}{\partial R} \frac{dR}{dt} + \frac{\partial L}{\partial u} \frac{du}{dt} = \frac{dR}{dt} (2R\Omega + u) + R \frac{du}{dt} = 0$$

so that,

$$\frac{du}{dt} = -\frac{(2R\Omega + u)}{R} \frac{dR}{dt} = -2\Omega \frac{dR}{dt} - \frac{u}{R} \frac{dR}{dt}$$

By geometry, $\Delta R = \Delta z \cos \phi$, $R = a \cos \phi$

$$\frac{du}{dt} = -2\Omega \frac{dz \cos \phi}{dt} - \frac{u}{R} \frac{dz \cos \phi}{dt} = -2\Omega w \cos \phi - \frac{uw}{a}$$

- (3) We have derived hypsometric equation and demonstrated the pressure profile of a continental cold high. In a similar fashion, please show the vertical pressure profile of an ideal tropical cyclone and Pacific high

- (4) Show that the transformation of the total derivative for a vector between inertial reference frame and rotating frame is given by

$$\frac{d_a A}{dt} = \frac{dA}{dt} + \Omega \times A$$

Where Ω is angular velocity and A is an arbitrary vector whose Cartesian components in an inertial frame are given by

$$A = i' A'_x + j' A'_y + k' A'_z$$

and whose components in a rotating frame are given

$$A = i A_x + j A_y + k A_z$$

Let $\frac{d_a A}{dt}$ be a total derivative of A in the initial frame

$$\begin{aligned} \frac{d_a A}{dt} &= i' \frac{dA'_x}{dt} + j' \frac{dA'_y}{dt} + k' \frac{dA'_z}{dt} \\ &= i \frac{dA_x}{dt} + j \frac{dA_y}{dt} + k \frac{dA_z}{dt} + \frac{d_a i}{dt} A_x + \frac{d_a j}{dt} A_y + \frac{d_a k}{dt} A_z \end{aligned}$$

in rotating coordinate,

$$\frac{dA}{dt} = i \frac{dA_x}{dt} + j \frac{dA_y}{dt} + k \frac{dA_z}{dt}$$

Consider eastward direct,

$$\delta i = \frac{\partial i}{\partial \lambda} \delta \lambda + \frac{\partial i}{\partial \phi} \delta \phi + \frac{\partial i}{\partial z} \delta z$$

for solid-body rotation

$$\delta \lambda = \Omega \delta t, \quad \delta \phi = 0, \quad \delta z = 0$$

$$\Rightarrow \frac{\delta i}{\delta t} = \frac{\partial i}{\partial \lambda} \frac{\delta \lambda}{\delta t}$$

take $\delta t \rightarrow 0$

$$\frac{d_a i}{dt} = \Omega \frac{\partial i}{\partial \lambda}$$

longitudinal derivative of i

$$\frac{\partial i}{\partial \lambda} = j \sin \phi - k \cos \phi$$

$$\Omega = (0, \Omega \cos \phi, \Omega \sin \phi) \Rightarrow \frac{d_a i}{dt} = \Omega(j \sin \phi - k \cos \phi) = \Omega \times i$$

Similarly, $\frac{d_a j}{dt} = \Omega \times j$ and $\frac{d_a k}{dt} = \Omega \times k \rightarrow \frac{d_a A}{dt} = \frac{dA}{dt} + \Omega \times A$

- (5) Considering the directions of the i, j, k unit vectors are not constant in spherical coordinates, the total derivatives of velocity can be expressed

$$\frac{dV}{dt} = \frac{du}{dt} i + \frac{dv}{dt} j + \frac{dw}{dt} k + u \frac{di}{dt} + v \frac{dj}{dt} + w \frac{dk}{dt}$$

Now please expand the total derivatives of the last two terms in the right-hand side into their local derivatives and advection parts. Explain their physical meanings.

Ans:

With the following figure eastward motion : $|\delta j| = \frac{\delta x}{(a/\tan \phi)}$

Because the vector $\frac{\partial j}{\partial x}$ is directed in the negative x direction, we then have

$$\frac{\partial j}{\partial x} = -\frac{\tan \phi}{a} \vec{i}$$

northward motion $|\delta \vec{j}| = \delta \phi$, but $\delta y = a \delta \phi$, and δj is directed downward so that

$$\frac{\partial \vec{j}}{\partial y} = -\frac{\vec{k}}{a}$$

Hence,

$$\frac{d\vec{j}}{dt} = -\frac{-u \tan \phi}{a} \vec{i} - \frac{v}{a} \vec{k}$$

Similarly, the vector $\delta k = \frac{\delta x}{a}$ is directed in the positive x direction, thus

$$\frac{\partial k}{\partial x} = \frac{\vec{i}}{a}$$

the vector $\delta k = \frac{\delta y}{a}$ is directed upward, thus

$$\frac{\partial \vec{k}}{\partial y} = \frac{\vec{j}}{a}$$

Hence,

$$\frac{d\vec{k}}{dt} = \frac{u}{a}\vec{i} + \frac{v}{a}\vec{j}$$

In summary,

$$\begin{aligned} v \frac{d\vec{j}}{dt} &= v \left(-\frac{u \tan \phi}{a} \vec{i} - \frac{v}{a} \vec{k} \right) = -\frac{uv \tan \phi}{a} \vec{i} - \frac{v^2}{a} \vec{k} \\ w \frac{d\vec{k}}{dt} &= w \left(\frac{u}{a} \vec{i} + \frac{v}{a} \vec{j} \right) = \frac{uw}{a} \vec{i} + \frac{vw}{a} \vec{j} \end{aligned}$$